Lovely Ram

India | +91 79820 57996 | lovelyrontech@gmail.com | linkedin.com/in/lovely-ram-62730b36

Engineering Leader · Architect — Cloud-Native Data & SaaS Platforms, AI Knowledge Systems

IIT Hyderabad Certified — Professional Certification Program in AI & Emerging Technologies (2026)

EXECUTIVE SUMMARY

Software engineering leader and architect with **14+ years** building secure, scalable, data-intensive enterprise platforms, including **5 years** in tech lead roles leading and mentoring engineering teams across architecture, code quality, testing, and delivery. I own the full lifecycle — from requirements and system design to cloud-native deployment on **AWS** and **Azure** with **Docker**, **Kubernetes**, and **GitHub Actions** CI/CD — and build in **Python** and **Java**. I have delivered event-driven microservices on **Apache Kafka** for high-volume retail data flows, workflows, APIs, automation, and observability on the **ServiceNow** SaaS enterprise platform, a **multi-tenant SaaS** payments platform serving merchants and ISVs, and production AI knowledge systems: a central **Agentic AI Orchestrator** and reusable **MCP**-based framework adopted as a shared capability across multiple use cases, a chatbot integrated with **Databricks Genie** over Lakehouse data, and **RAG** and semantic-search systems over enterprise knowledge. I design systems to be secure, auditable, and observable — guardrails, validation, controlled tool access, deterministic logic beneath AI layers — and led vulnerability remediation across **90+ microservices**. I back this with delivery depth: migrating data from a **24-year-old legacy system** through phased testing, reconciliation, and a runbook-driven go-live, and cutting latency by **~40–50%**. I translate architecture into business terms for product, business, and compliance stakeholders. IIT Hyderabad-certified in AI & Emerging Technologies. Domain expertise spans retail merchandising, insurance (life, group, and general), and payments.

CORE ENGINEERING LEADERSHIP & TECHNOLOGY COMPETENCIES

Engineering Leadership & Team Building • Software & Quality Engineering Programs • Engineering Roadmaps with Product • End-to-End SDLC Ownership • Cloud-Native Architecture (AWS, Azure) • SaaS & Multi-Tenant Platform Architecture • SaaS Platform Integration (ServiceNow) • Containers & Kubernetes • CI/CD & DevSecOps • Event-Driven & High-Volume Data Systems • Data Platforms, Migration & Reconciliation • Semantic Search, RAG & Knowledge Retrieval • Agentic AI & MCP Integration • Rapid Prototyping & R&D • Reusable Shared Capabilities • Security, Compliance, Auditability & Observability • Production Go-Live & Runbook Management • Stakeholder Communication (Product, Business, Compliance) • Agile Delivery & Change Management

DOMAIN EXPERTISE

Retail & Merchandising (Adidas): Market/Range Management across Range, Segmentation, Pricing, Volume, Speed/Size, and Margin domains; range data aggregation and downstream reporting/analytics.

Insurance (DXC Technology, Diaspark): Life, group, and general insurance; policy administration on CSC's Integral™ suite — product configuration, policy and group management, multi-currency, multi-language, multi-company/multi-branch processing, and country-specific regulatory configuration.

Payments (Capita): Pay360 multi-tenant SaaS online payment platform — secure payment APIs, authentication/authorization, tenant-scoped (merchant, ISV, operations) permissions, and payment-service integration.

TECHNICAL SKILLS

Languages & Architecture: Python, Java 8–21 (incl. JDK 17 & 21), Spring Boot, JPA, Hibernate, REST APIs, Microservices, Distributed Systems, Event-Driven Architecture, SaaS & Multi-Tenant Architecture, Enterprise/Solution Architecture, System Design

SaaS Platforms: ServiceNow — Workflow Development, Integration APIs, Automation, Observability; Multi-Tenant SaaS Architecture

Cloud & DevOps: AWS, Azure, Docker, Kubernetes, CI/CD, GitHub Actions, Maven, Git, Vulnerability Management

Observability: Prometheus, Grafana, Monitoring, Alerting

Data Platforms & Streaming: Apache Kafka, Databricks, Lakehouse, Data Aggregation, Data Migration & Reconciliation, Data Archival & Purge, Oracle, MS SQL Server, PostgreSQL, MongoDB

Semantic, Generative & Agentic AI: Semantic Search, Vector Embeddings, Vector Databases, RAG, Enterprise Search, Knowledge Retrieval, Agentic AI, Multi-Tool Agent Orchestration, MCP, FastMCP, LangGraph, Guardrails, Prompt & Context Engineering, Hybrid Rule-Based & LLM Automation, Ollama, OpenAI API, Claude AI, Gemini API, AWS Bedrock, AWS SageMaker, Databricks Genie, GitHub Copilot

Quality Engineering: Phased Migration Testing, Performance Testing, End-to-End Testing, Data Reconciliation, Quality Gates, Code Review Standards, Production Go-Live Readiness, Runbook Management

Frontend & Web: JavaScript, TypeScript, HTML, CSS, jQuery

Security & Delivery: JWT, OAuth2, Authentication & Authorization, ADRs, NFRs, Agile, SAFe

PROFESSIONAL EXPERIENCE

Senior Software Engineer / Tech Lead — Solution Architect (AI & Agentic AI Initiatives) **2023 – Present**

Adidas India Pvt Ltd

- Architected and delivered a central **Agentic AI Orchestrator** for the MRM/M3 Retail Domain that interprets user intent, defines agent task flows with human-in-the-loop clarification when information is missing, and routes only validated requests to domain-bounded **MCP** servers — establishing a reusable agentic framework consumed as a shared capability across multiple use cases.
- Defined controlled tool access and bounded autonomy for agents: validation, guardrails, timeout/error handling, and fallback behaviour, separating LLM reasoning from enterprise execution so consequential actions stay secure, auditable, and reliable.
- Designed M3 Support Automation as a hybrid rule-based/LLM detection engine on **Kafka** — deterministic rules where they suffice, LLMs only where judgment is needed; introduced local **Ollama** execution to cut external-dependency risk and inference cost.
- Integrated the M3 Merchandise chatbot with multiple MCP tools, including a **Databricks Genie** tool over Lakehouse data, giving business users governed natural-language access to merchandise and operational data.
- Built an AI incident-management copilot using **RAG** over historical logs, runbooks, and internal documentation — classifying incidents SEV-1 to SEV-4 and recommending self-healing actions for engineers to verify, reducing manual triage effort for the support team.
- Developed and integrated workflows, APIs, automation, and observability on **ServiceNow**, a **SaaS**-based enterprise platform.
- Lead technologist for Generative AI and Agentic AI initiatives on the M3 platform, partnering with business teams to scope requirements, prototype, and prioritize agent, copilot, and automation use cases by business value.
- **M3 MRM – Market/Range Management:** Contributed to a multi-product, event-driven, high-volume data platform supporting Range, Segmentation, Pricing, Volume, Speed/Size, and Margin domains — building microservices, Apache Kafka flows, DAL/data aggregation, REST APIs, and downstream integrations, including single/bulk update and status-change workflows feeding BWCL and Range Insights for reporting and analytics.

  Architecture: *Frontend → Market Ranging → Domain Microservices → Kafka → DAL/Data Aggregation → Aggregated Kafka Topic → BWCL/Range Insights*
- Own end-to-end solution architecture across AI and platform initiatives (Support Automation, Data Archival & Purge, Log Analysis, Agentic Orchestrator) on a large-scale distributed platform — defining component boundaries, integration contracts, NFRs, and ADRs, evaluating architecture alternatives and trade-offs, leading design reviews, and mentoring engineers building on these platforms.
- Played a key role in decommissioning RMA, a **24-year-old legacy system**, and building its replacement, the M3 application — validating migrated data through phased migration testing, data reconciliation, performance testing, and end-to-end testing before cut-over.
- Core member of the M3 production go-live team — authored, owned, and executed go-live runbooks covering cut-over steps, validation checkpoints, and rollback procedures.
- Partnered with Security teams to lead vulnerability remediation across **90+ microservices**, drove production go-live readiness, and reduced end-to-end response latency by **~40–50%** through JVM/concurrency optimization; resolved a critical payment API timeout in ~25 minutes during production support.

Senior Software Engineer / Tech Lead — Solution Architect **2021 – 2023**

Capita India Pvt Ltd

- Led a team of **5 engineers** delivering Pay360, Capita's microservices-based, **multi-tenant SaaS** online payments platform — owning architecture, engineering standards, code quality, testing, and delivery end to end.
- Designed secure APIs and authentication/authorization (**JWT**, **OAuth2**) with tenant-scoped permissions at merchant, ISV, and operations level, and **Kafka**-based integration to decouple services and improve throughput and resilience.
- Defined solution architecture and engineering standards that improved delivery efficiency by **~20%**; stronger development and code-review practices reduced defects and rework by **~40–50%**.

Product Developer **2014 – 2021**

DXC Technology

- Built and enhanced life, group, and general insurance solutions on CSC's Integral™, an end-to-end policy administration suite covering all insurance lines — including product configuration, policy/group management, and business analytics.
- Worked across Integral's global capabilities: foreign-currency processing, multi-language support, multi-company and multi-branch processing, and regional configuration for dates, addresses, and country-specific regulatory norms.

- Developed on ProductStage and the ITS microservice, using **Java**, Spring, REST APIs, JavaScript, jQuery, HTML, and CSS — contributing to solution design, technical feasibility, legacy modernization, and enterprise/downstream system integration.

Software Engineer **2013 – 2014**

Diaspark Pvt Ltd

- Developed and maintained enterprise insurance applications using **Java**, Spring, and REST APIs; contributed to technical design, feasibility analysis, and service-oriented architecture.

Software Engineer **2012 – 2013**

Datamatics Pvt Ltd (Honeywell Technology)

- Developed features for Honeywell's internal Business Management System (BMS) using **Java** and Spring; performed development, unit testing, debugging, and defect resolution.

SELECTED AI & ENGINEERING IMPACT

- Established an enterprise Agentic AI capability from the ground up — orchestrator + reusable MCP-based framework with bounded tool access, adopted across the M3 Retail Domain
- Helped retire a **24-year-old legacy system** (RMA) and migrate its data to M3 through phased testing, reconciliation, and a runbook-driven production go-live
- Built a RAG-based AI incident copilot with automated SEV classification and self-healing recommendations for IT operations
- Delivered measurable outcomes: **~40–50%** latency reduction, **~20%** delivery-efficiency improvement, **~40–50%** reduction in defects/rework
- Led vulnerability remediation across **90+ microservices** in partnership with Security teams
- Led and mentored engineering teams (**5+ engineers**) across architecture, delivery, quality, and secure development

INDEPENDENT PROJECTS & RAPID PROTOTYPES

Fund Kuber — AI-Powered Platforms for a Mutual Fund Distributor **2025 – Present**

- **Portfolio & Risk Analysis Platform:** Architecting a full-stack platform covering client onboarding, risk profiling, portfolio-statement ingestion/extraction, analytics, goal simulation, and AI chat, with deterministic financial calculations beneath an AI explanation layer to keep outputs auditable and compliance-safe.
- **Marketing Automation AI Agent:** Built an AI marketing platform that analyzes social-media engagement to surface content and audience trends, generates brand-aligned post copy with LLMs, validates it before publishing via the Facebook Graph and Instagram APIs, and closes the loop with performance tracking (Social Data → Engagement Analysis → AI Insights → LLM Content Generation → Validation → Scheduling/Publishing → Performance Tracking) — reducing manual effort and designed as a reusable agent extensible to new channels.

Telecom 5G Knowledge Assistant — RAG-Based Q&A System(Complete)

- Built a semantic Q&A system over 5G technical specifications: chunking, dense vector embeddings in a vector database, semantic search/retrieval, and LLM-based, source-grounded answer generation.

FundKuber Social Agent — AI-Powered SaaS Automation Platform

- Architected an AI-driven social automation platform using a **multi-stage agentic workflow** for content planning, research, copywriting, creative generation, compliance, and multi-platform publishing.
- Implemented **AI provider orchestration, retries and fallback strategi**es to handle quota, rate-limit, model and provider failures.
- Integrated Facebook, Instagram and LinkedIn APIs with platform-specific publishing workflows.
- Built **end-to-end observabilit**y covering scheduled/manual executions, stage-level latency, AI inference, provider failures, retries, fallbacks and actionable root-cause diagnostics.
- Implemented content-quality controls including **duplicate detection, content-history tracking, compliance validation and automated disclaimers.**
- Designed persistent execution history and artifact management to support **reliable, diagnosable automation workflows.**

NotePicker — Chrome Extension

- Designed and built a Manifest V3 note-taking/highlighting extension using JavaScript, TypeScript, HTML, and CSS, from PRD through implementation.

CERTIFICATIONS & EDUCATION

- Professional Certification Program in AI & Emerging Technologies — IIT Hyderabad, 2026
- SAFe (Scaled Agile Framework) Certified
- Master of Computer Applications (MCA), Biju Patnaik University of Technology (BPUT), Odisha, 2011